Ericsson PE Node (BAS-PE-10) Hardware/Environmental Fault Alarm Handling (ALARM_7837)

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_7837

Description

The Ericsson PE node (BAS-PE-10) at the Basavanagudi POP may experience hardware/environmental faults leading to ASIC memory parity errors, line card crashes, or power supply redundancy loss, resulting in the ALARM_7837 warning. This failure domain occurs on this specific role due to the inherent design of the Ericsson PE node, which utilizes a distributed architecture with multiple line cards and power supplies to provide high availability. However, in the event of a hardware failure, the distributed architecture can exacerbate the issue, leading to a cascading effect on the control and data planes. The ASIC memory parity errors and line card crashes can be attributed to issues with the high-speed interfaces, such as the 100GE or 400GE interfaces, which are prone to errors due to factors like electromagnetic interference (EMI), temperature fluctuations, or software bugs. The power supply redundancy loss can be caused by a faulty power supply unit (PSU) or a misconfiguration of the power redundancy mechanism. The ALARM_7837 warning is triggered when the system detects a critical failure in the hardware or environment, indicating the need for immediate attention to prevent further damage or downtime.

Pre-requisite Checks

- Item: Verify the power supply redundancy status using the command "admin show redundancy power-supply", Description: Ensure that the power supplies are properly configured and functioning correctly.

- Item: Check the status of the line cards using the command "admin show line-card", Description: Verify that all line cards are operational and not experiencing any issues.

- Item: Review the system logs using the command "admin show log", Description: Analyze the system logs to identify any patterns or anomalies that may indicate the root cause of the failure.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: admin show hardware asic-memory-parity-errors, Expected Output: ASIC memory parity errors detected on one or more line cards, Description: This command displays the status of the ASIC memory parity errors, indicating whether any errors have been detected on the line cards.

Command: admin show line-card status, Expected Output: Line card status: active/standby, Description: This command displays the status of the line cards, indicating whether they are active or in standby mode.

Command: admin show redundancy power-supply status, Expected Output: Power supply status: active/standby, Description: This command displays the status of the power supplies, indicating whether they are active or in standby mode.
```

Resolution Steps

Step 1: Step: Replace faulty line card, Command: admin replace line-card , Description: Replace the faulty line card with a new one, and ensure that the redundant line card takes over the failed line card's functions.

Step 2: Step: Replace faulty power supply, Command: admin replace power-supply , Description: Replace the faulty power supply with a new one, and ensure that the redundant power supply takes over the failed power supply's functions.

Step 3: Step: Reset the ASIC memory parity errors, Command: admin reset asic-memory-parity-errors, Description: Reset the ASIC memory parity errors, and ensure that the system recovers from the failure.

Post-Check Verification

```
Command: admin show hardware asic-memory-parity-errors, Expected Output: No ASIC memory parity errors detected, Description: Verify that the ASIC memory parity errors have been resolved.

Command: admin show line-card status, Expected Output: Line card status: active, Description: Verify that the line card has been successfully replaced and is now active.

Command: admin show redundancy power-supply status, Expected Output: Power supply status: active, Description: Verify that the power supply has been successfully replaced and is now active.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: admin restore configuration, Description: Restore the original configuration to ensure that the system returns to its previous state.

- Step: Replace the faulty component, Command: admin replace , Description: Replace the faulty component with a new one, and ensure that the system recovers from the failure.